Cover illustration: 31m American type purse seiner – General arrangement.
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Classification and definition of fishing vessel types

Second edition

FAO
FISHERIES AND
AQUACULTURE
TECHNICAL
PAPER

267

Stefano Thermes
International consultant

Raymon van Anrooy
Food and Agriculture Organization of the United Nations

Ari Gudmundsson
International consultant

and

Daniel Davy
International consultant

FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS
Rome, 2023

Required citation:
Thermes, S., Van Anrooy, R., Gudmundsson, A. & Davy, D. 2023. *Classification and definition of fishing vessel types.* Second edition. FAO Fisheries and Aquaculture Technical Paper, No. 267. Rome, FAO. https://doi.org/10.4060/cc7468en

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ISSN 2070-7010 [Print]
ISSN 2664-5408 [Online]

ISBN 978-92-5-138104-5
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Preparation of this document

This document is an updated version of FAO Fisheries Technical Paper 267, *Definition and classification of fishery vessel types*, published in 1985.

There are two reasons for producing this second edition. The first is the design, size and other characteristics of the main types of semi-industrial and industrial fishing vessels, which have changed significantly in the intervening decades since the original publication. The second reason was the revision of the International Standard Statistical Classification of Fishery Vessels by Vessel Types (ISSCFV), which was reviewed and endorsed at the Twenty-sixth Session of the Coordinating Working Party on Fishery Statistics (CWP), held in Rome, Italy, from 15 to 18 May 2019.

The document provides illustrations of most vessel types, as well as drawings of the general arrangements of many vessels, but also includes several important updates with respect to the 1985 edition. Most notably, the vessel type classification codes employed have been updated, as has the approximate relationship between vessel length overall (LOA) and gross tonnage (GT), based on global data from IHS Markit. The present edition also includes an extensive glossary of commonly used terminology in the description of fishing vessels and their operations.

The text was written in 2020–2021 by Stefano Thermes, Naval Architect, supported by Daniel Davy, Naval Architect, Ari Gudmundsson, Naval Architect, and Raymon van Anrooy of the FAO Fisheries and Aquaculture Division. The general arrangement drawings and related vessel pictures included were provided by various shipyards and naval architects. The 3D vessel illustrations were procured by Navalia – marine design software specialists. The draft document was reviewed by naval architects and FAO fisheries statistics officers.

The publication was edited by Edward Fortes, with formatting and design assistance was provided by Chorouk Benkabbour and Marianne Guyonnet of the FAO Fisheries and Aquaculture Division.

Abstract

This technical paper provides descriptions and classification of the main semi-industrial and industrial fishing vessel types: trawlers, purse seiners, seiners, dredgers, gillnetters, trap setters, longliners, pole-and-line vessels, trollers, and multipurpose vessels. It also covers vessels supporting fishing-related activities such as fish carriers, motherships, fisheries research vessels and vessels involved in aquaculture operations. It includes illustrations of most of the vessel types and drawings of the general arrangements of selected vessels. The design, size and other characteristics of the main types of semi-industrial and industrial fishing vessels have changed significantly in recent decades, so the second edition includes several important updates with respect to the 1985 version. The vessel type classification codes employed have been updated, as has the approximate relationship between fishing vessels' length overall (LOA) and their gross tonnage (GT).

The classification of the fishing vessel types follows the International Standard Statistical Classification of Fishery Vessels by Vessel Types (ISSCFV) that was endorsed by the Coordinating Working Party on Fishery Statistics (CWP) in 2019. The descriptions and classification are valid worldwide.

The main purpose of this technical paper is to assist FAO Members, regional fishery bodies, as well as those working on fishery statistics and management, with updated information on vessel types and characteristics. It provides users, including non-specialists, with sufficient information to identify and classify all types of semi-industrial and industrial fishery vessels for reporting purposes.

The paper also contributes to FAO's work in support of the implementation of the 1999 International Plan of Action for the Management of Fishing Capacity (IPOA-Capacity), with regard to the prevention, deterrence and elimination of illegal, unreported and unregulated (IUU) fishing. It does so by providing monitoring, control and surveillance personnel with information to identify the type of fishery vessels in terms of their licensing and authorization to carry out fishing and fishing-related operations. The paper also complements the International Standard Statistical Classification of Fishing Gear (ISSCFG) and the FAO publication *Classification and illustrated definition of fishing gears*.

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Acknowledgements

The authors would like to acknowledge the support received from the following ship design offices, shipyards and industry specialists, which contributed to this paper with technical documentation, vessel illustrations and information on vessel particulars:

Damen Maaskant Shipyards Stellendam
Stellendam, Kingdom of the Netherlands
www.damen.com

European Commission
Directorate-General for Maritime Affairs and Fisheries
Brussels, Belgium
https://ec.europa.eu/info/departments/maritime-affairs-and-fisheries_en

IHS Markit
London, United Kingdom of Great Britain and Northern Ireland
https://ihsmarkit.com/industry/maritime.html

Kooiman Marine Group
Zwijndrecht, Kingdom of the Netherlands
www.kooimanmarinegroup.nl

NAVALIA
Naples, Italy
info@navalia.org

NAVIS ehf
Reykjavik, Iceland
www.navis.is

Selfa Arctic AS
Sandtorg, Norway
www.selfa.no

Skipasyn-Icelandic
Reykjavik, Iceland
www.skipasyn.is

Skipsteknisk A/S
Aalesund, Norway
www.skipsteknisk.no

Studio Navale Samarelli
Molfetta, Italy
www.samarellistudionavale.com

The authors would also like to express their gratitude to: Derrick Menezes, Naval Architect (FAO consultant), Liming Song (Shanghai Ocean University), Rajdeep Mukherjee (Bay of Bengal Programme Inter-Governmental Organization), Isara Chanrachkij (Southeast Asian Fisheries Development Center), Stefania Vannuccini, Jennifer Gee, José Acuna, Marc Taconet, Jon Lansley and Aureliano Gentile from the FAO Fisheries and Aquaculture Division (NFI). We are grateful to these individuals for their advice on the International Standard Statistical Classification of Fishery Vessels (ISSCFV), the sharing of data and information from the Global Record of Fishing Vessels, Refrigerated Transport Vessels and Supply Vessels (Global Record) and the FAO fishing fleet statistical database, as well as for reviewing this publication.

Acronyms and abbreviations

CWP	Coordinating Working Party on Fishery Statistics (FAO)
EEZ	exclusive economic zone
FAD	fish aggregating device
FAO	Food and Agriculture Organization of the United Nations
GHG	greenhouse gas
GPS	global positioning system
Global Record	Global Record Fishing Vessels, Refrigerated Transport Vessels and Supply Vessels (FAO)
ILO	International Labour Organization
IMO	International Maritime Organization
IPOA	international plan of action
ISSCFG	International Standard Statistical Classification of Fishing Gear
ISSCFV	International Standard Statistical Classification of Fishery Vessels
IUU	illegal, unreported and unregulated fishing
LOA	length overall
NEI	not elsewhere included
PSMA	Agreement on Port State Measures to Prevent, Deter and Eliminate Illegal, Unreported and Unregulated Fishing
RFMO	regional fisheries management organization
RSW	refrigerated sea water

1. Introduction

The global fishing industry is dynamic: fishing fleets have to adjust to economic, financial, technological, social, and environmental developments. The financial and economic performance of fishing fleet operations is thus affected by many factors: the status of fisheries resources, fisheries policies, legislation and management regimes, seafood market prices and consumer preferences, macroeconomic and fiscal policies, fisheries agreements, technological innovations and socioeconomic conditions.

The fishing vessels which make up the semi-industrial and industrial fishing fleets have changed significantly in recent decades. Vessel and gear technologies, as well as innovations in fishing vessel design, propulsion systems, onboard equipment and fishing methods, have impacted the characteristics of fishing vessels and their operations. Similarly, the vessels that have entered the semi-industrial and industrial fishing fleets worldwide are generally larger in length overall (LOA), gross tonnage (GT) and engine power (in kilowatts) than in the past (Van Anrooy *et al.*, 2021). The average fishing vessel size, the number of fishing vessels and the overall fishing fleet capacity to catch fisheries resources are all greater today than at the start of the twenty-first century.

The increase in vessel size and fishing capacity has been largely driven by profitability goals and technological innovation, as well as national legislation and international conventions on the safety of fishing vessels, and the safety and working conditions of fishing vessel personnel.

Any assessment of fishing fleet capacity requires, as a bare minimum, estimates of fishery vessel numbers and main vessel characteristics. If the fleet consists of only one type of vessel, the number of fishing vessels may be used to express the total fishing power (or "capacity") of the fishing fleet. If the fleet consists of vessels of different types, any survey to determine the capacity of a given fishing vessel would need to collect information on a number of vessel characteristics including vessel type, gross tonnage, length, and engine power. The exact fishing capacity indicator used will depend on the characteristics of the fishery or fleet, and the availability of reliable data. For example, it is generally accepted that the single most important factor for trawlers is engine power. For gillnetters, on the other hand, engine power would be of limited importance. Applying a universal capacity measure across a range of fisheries may therefore prove inadequate when addressing the issue of global fishing capacity measurement (FAO, 2023c).

This technical paper provides descriptions and classification of the main semi-industrial and industrial fishing vessel types: trawlers, purse seiners, seiners, dredgers, gillnetters, trap setters, longliners, pole-and-line vessels, trollers, and multipurpose vessels. It also includes vessels supporting fishing-related activities such as fish carriers, motherships, fisheries research vessels and vessels involved in aquaculture operations. Illustrations of most of the vessel types and drawings of the general arrangements of selected vessels are also provided.

The main purpose of this paper is to assist FAO Members, regional fishery bodies, as well as those working on fishery statistics and management with updated information on vessel types and characteristics. It provides users, including non-specialists, with sufficient information to identify and classify all types of semi-industrial and industrial fishery vessels clearly for reporting purposes .

The paper constitutes an update on the FAO publication *Definition and classification of fishery vessel types* (FAO, 1985) and incorporates the revised (2019)

fishery vessel type codes from the International Standard Statistical Classification of Fishery Vessels by Vessel Types (ISSCFV), as endorsed by the Twenty-sixth Session of the Coordinating Working Party on Fishery Statistics (FAO, 2019). The focus of this paper is on decked fishing vessels of 18 metres in length and over (ISSCFV Length Classes code 120 and higher).

The paper also contributes to the implementation of the 1999 International Plan of Action for the Management of Fishing Capacity (IPOA-Capacity) (FAO, 1999), and to the prevention, deterrence and elimination of illegal, unreported and unregulated (IUU) fishing. It does so by providing monitoring, control and surveillance personnel with information to identify the type of fishery vessels in terms of their licensing and authorization to carry out fishing and fishing-related operations.

FAO collects fishing fleet information from its Members on an annual basis, using statistics reporting forms agreed by the Coordinating Working Party on Fishery Statistics (FAO, 2021). The data collected include the number of decked vessels and their total gross tonnage (GT), total engine power (kW) by length overall (LOA) class and by fishing vessel type, as well as the number of undecked, powered and unpowered vessels, by LOA class and fishing vessel type. The received data are published in the FAO Yearbook on Fishery and Aquaculture Statistics (FAO, 2021a). The proper identification of fishing vessel types is therefore essential for statistical reporting on fishery fleets.

The descriptions of the types of fishery vessel provided in this technical paper, as well as those of vessels that support fishing-related activities, cover most of these vessels used worldwide. Users should note that vessel type descriptions are simplified and indicative: while individual vessels may perform similar fishing operations and/or use similar fishing gears, they may have significantly different general arrangements and characteristics, including differences in GT, LOA, and kW.

The 2019 ISSCFV is included in Appendix 1, and the International Standard Statistical Classification of Fishery Vessels by Length Classes (ISSCFV – Length Classes) can be found in Appendix 1A. Appendix 2 contains illustrations of the definition of length (L) of a vessel. An approximate relationship between the length overall (LOA) of fishery vessels and their gross tonnage (GT) is provided in Appendix 3.

The main energy source of almost all semi-industrial and industrial fishing vessels is fossil fuel. Appendix 4 provides information on some options that could contribute to reducing the fisheries industry's greenhouse gas (GHG) emissions.

This technical paper refers to a variety of fishing gears and should therefore be read in conjunction with the FAO publication, *Classification and illustrated definition of fishing gears* (He *et al.*, 2021), as well as the International Standard Statistical Classification of Fishing Gears (ISSCFG, 2016), included here in Appendix 1B.

2. Description of the categories of fishery vessels

This chapter contains description of the major categories and subcategories of fishery vessels, together with the codes and standard abbreviations as per ISSCFV Rev. 1, 2019 (FAO, 2021c), contained in Appendix 1.

2.1 TRAWLERS (TO)

ISSCFV Code 1

Overview

Depending on the area of operation and trawl used, trawlers range in size from open boats powered by outboard engines to large freezer and factory trawlers, which can fish in the most distant waters. Trawling is carried out from very shallow waters up to a depth of 2 000 m. These deepwater fishing vessels have machinery of sufficient power to tow the gear at the appropriate trawling speed. This will vary according to the species sought, from 1.5 knots (2.8 km/h) for shrimp trawls to 6 knots (11.1 km/h) or more for pelagic trawls.

Deck arrangement

From the point of view of deck arrangement, a distinction can be made between three main types of trawlers: side trawlers, stern trawlers and outrigger trawlers.

Equipment

Deck equipment – There are various arrangements of the deck equipment possible, depending on the type of trawler. All trawlers have trawl winches for the handling and storage of towing warps. Gilson winches, net drums and other auxiliary winches are commonly installed to handle the gear and the catch. Pelagic trawlers may be equipped with fish pumps for emptying the codend.

Fish detection equipment – Typical fish detection equipment of trawlers includes a sonar, net sounder, various types of echo sounders and trawl control and monitoring equipment.

Fishing gear – Mainly bottom trawls (TB), beam trawls (TBB), midwater trawls (TM), semipelagic trawls (TSP), and trawls NEI (TX).

Catch handling and processing equipment – Trawlers generally keep fresh fish in ice or refrigerated seawater (RSW) and/or frozen fish in blocks or boxes. Onboard cold chains with processing lines for slaughtering, sorting, and chilling fresh fish, as well as glazing and freezing systems (including plate and blast freezers and individually quick frozen systems) are also commonly installed on industrial trawlers.

FIGURE 1
Pelagic Stern Trawler

2.1.1 Beam trawlers (BT)

ISSCFV Code 1.3

Overview

A beam trawler's basic equipment is its outriggers, on which the gear is towed. Beam trawlers are the most widely used type of fishing vessel for flat fish and shrimp trawling in shallow waters. A single trawl or a twin trawl can be towed on each side. Beam trawlers are normally medium-sized, high-powered vessels and tow the gear at speeds between 2.5 and 8 knots (4.6 to 14.8 km/h), depending on the species targeted. Beam trawlers are named after the outriggers, which are steel beams to which the nets are attached and which holds the nets open. The bottom trawls used are cone-shaped nets, which are towed on the seabed and designed to catch fish living on or near the seabed. Safety release systems are sometimes incorporated in the boom stays or winch brakes to prevent the vessel from capsizing should the gear become snagged. For safety (stability) reasons, the engine power of beam trawlers may be limited in some countries.

Otter trawlers (OT) generally also use outriggers, though not necessarily steel beams. They are named after the otter boards used in single-boat bottom trawls, which help keep the net in contact with the seabed and are mainly used in Asia. Pair trawlers (PT) generally tow the bottom trawl net between two boats.

Deck arrangement

Beam trawlers that use outriggers may have their superstructure positioned forward with the working deck aft, as is the frequently the case when shrimp are targeted. The towing winch will then be located just aft of the superstructure (Figure 2). Beam trawlers that employ trawls rigged with large beams may have the wheelhouse and living quarters at the aft part of the vessel, with the working deck amidships. This arrangement allows for easier stowing and handling of large beams. In this case the outriggers will be located in association with a mast or A-frame set amidships or forward, and the towing winch will be located in front of the superstructure. The principal characteristic of beam trawlers is the presence of beams which are usually fastened to the mast or at the foot of the mast and extend out over the sides of the vessel during fishing operations.

Equipment

Deck equipment – Warp winches with capstans are installed on the deck of the vessels. Catch handling is supported by sorting devices and washing machines.

Fish detection equipment – Fish detection equipment, such as multibeam fish finders are used extensively on board beam trawlers. Fish finders have become common components of integrated systems with marine radar, compass, ECDIS and GPS navigation systems.

Fishing gear – Twin bottom otter trawls (OTT), beam trawls (TBB); single-boat bottom otter trawls (OTB), trawls NEI (TX).

Catch handling and processing equipment – All.

FIGURE 2
Gulf shrimp beam trawler (outrigger)

FIGURE 3
41m Beam trawler
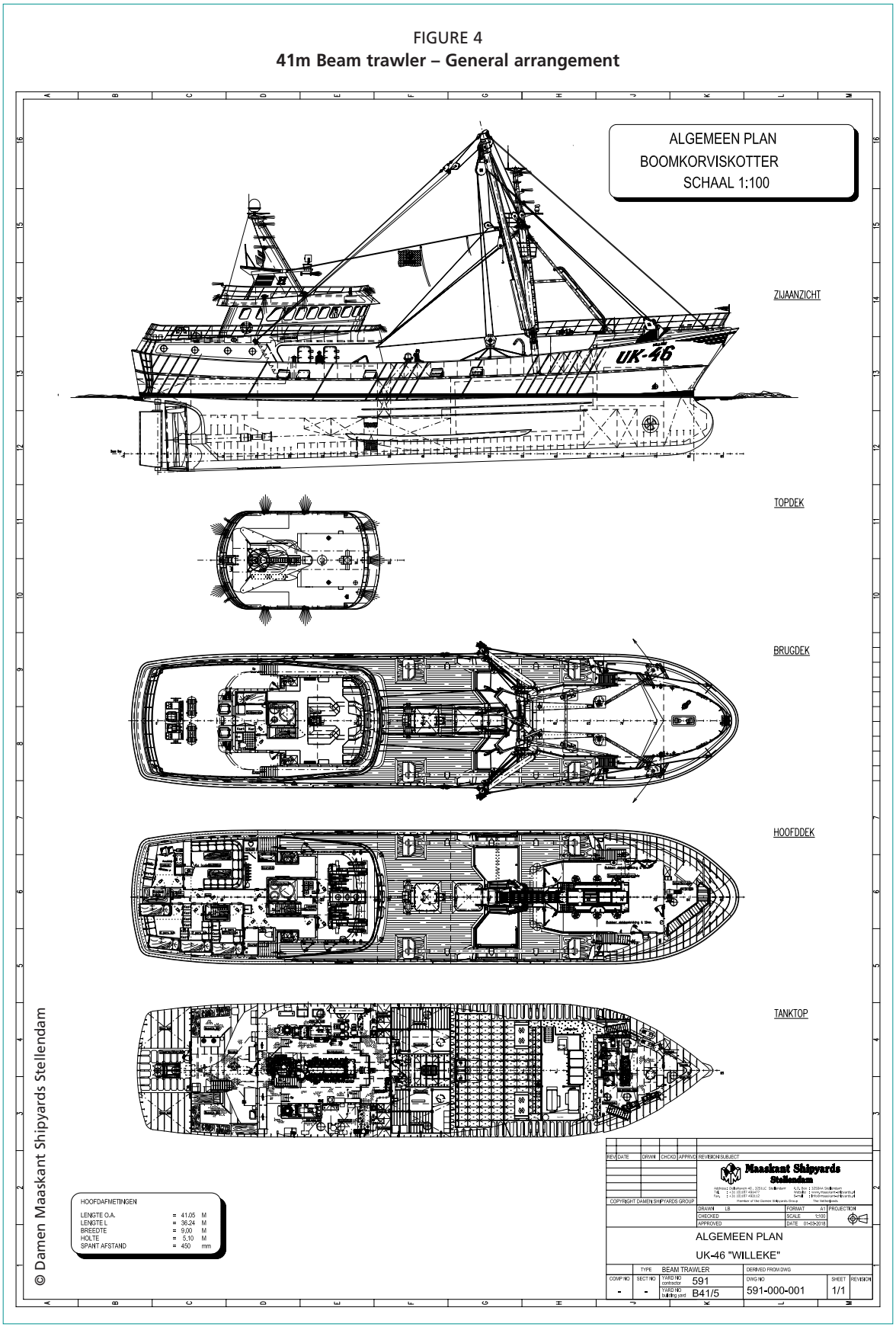
FIGURE 4
41m Beam trawler – General arrangement